

Supplementary file of “A Constrained Learning-Based Competitive Swarm Optimizer for Large-scale Multi-Objective Optimization”

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1. Supplementary Figures and Tables

1.1 Figures:

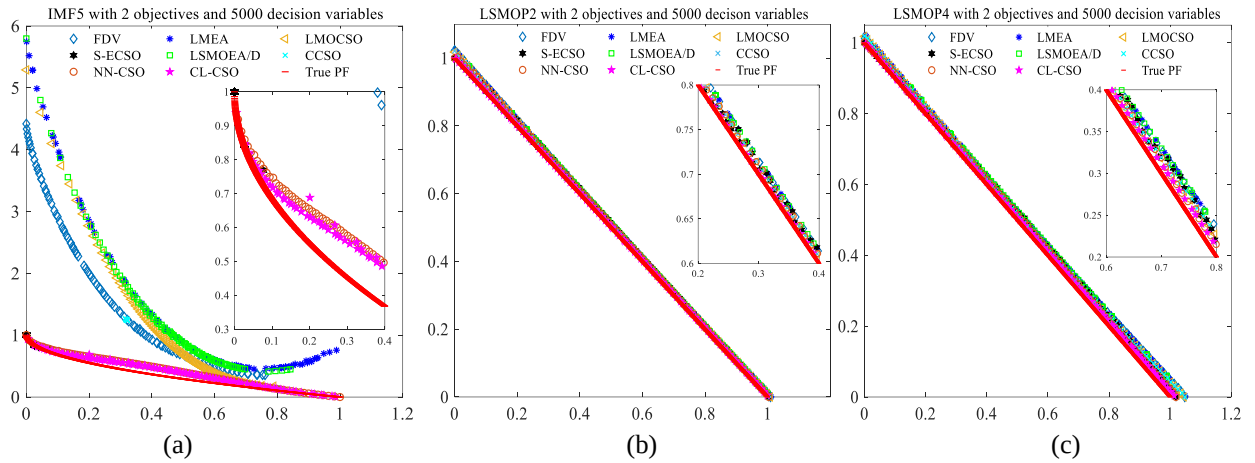


Fig. A. 1. The final population sets obtained by eight compared algorithms on a) 2-objective IMF5 with 5000 decision variables, b) 2-objective LSMOP2 with 5000 decision variables and c) 2-objective LSMOP4 with 5000 decision variables.

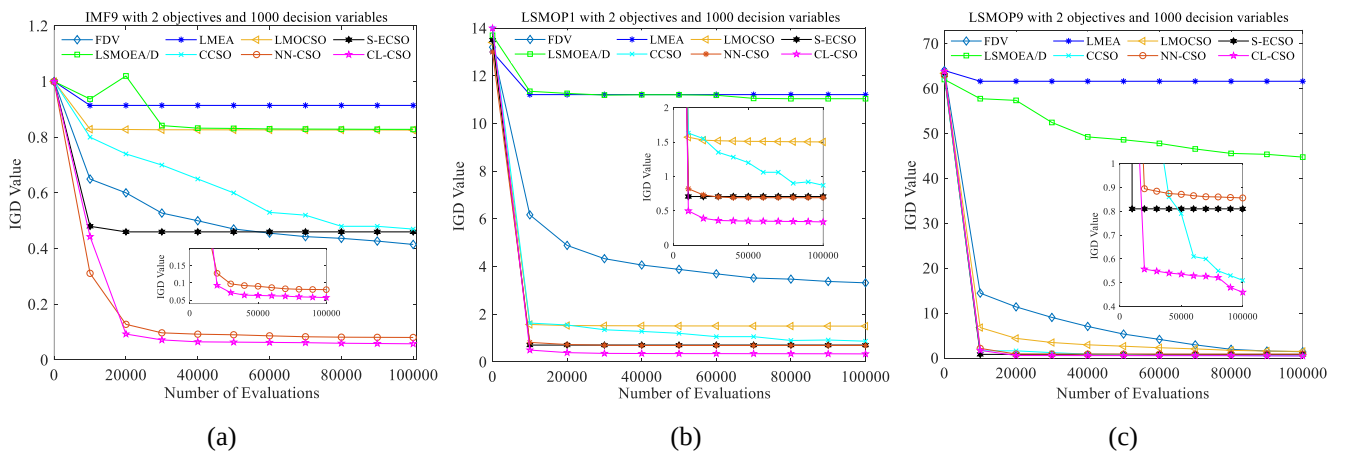


Fig. A. 2. The convergence profiles of eight compared algorithms on a) IMF9 with 2 objectives and 1000 decision variables, b) LSMOP1 with 2 objectives and 1000 decision variables, c) LSMOP9 with 2 objectives and 1000 decision variables, respectively.

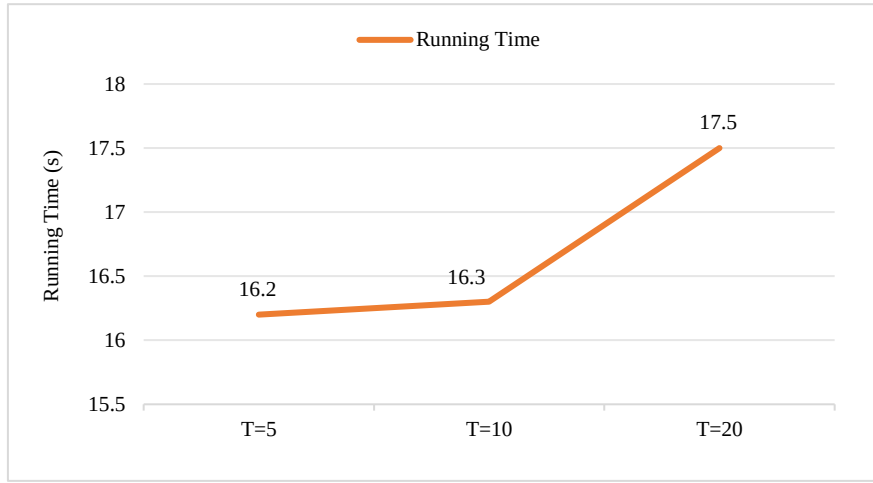


Fig. A. 3. Average running time of CL-CSO with different values of T in solving IMF1 problem with 500 decision variables

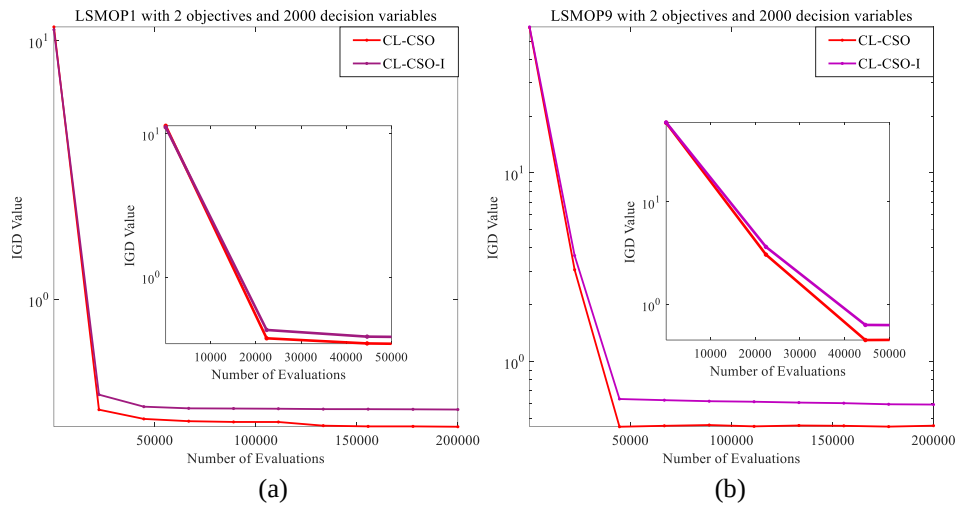


Fig. A. 4. The convergence profiles of CL-CSO and CL-CSO-I on a) 2-objective LSMOP1 with 2000 decision variables and b) 2-objective LSMOP9 with 2000 decision variables.

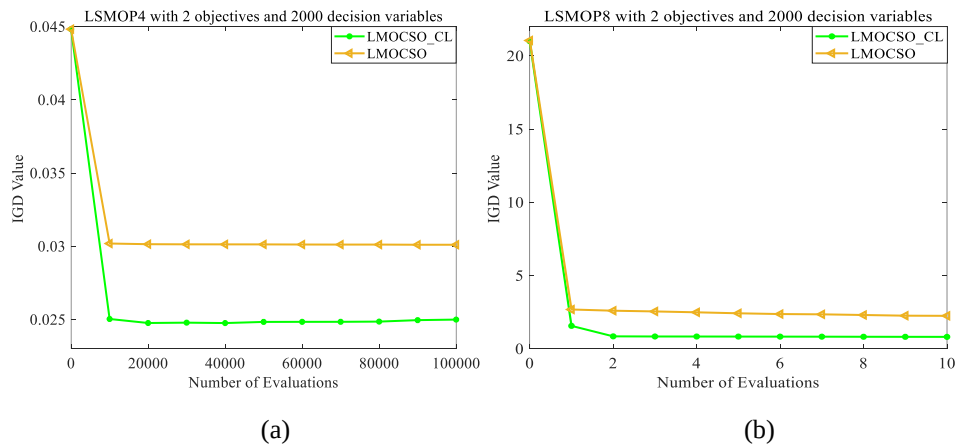


Fig. A. 5. Convergence profiles of LMOCSO and LMOCSO_CL on a) 2-objective LSMOP4 with 2000 decision variables and b) 2-objective LSMOP8 with 2000 decision variables.

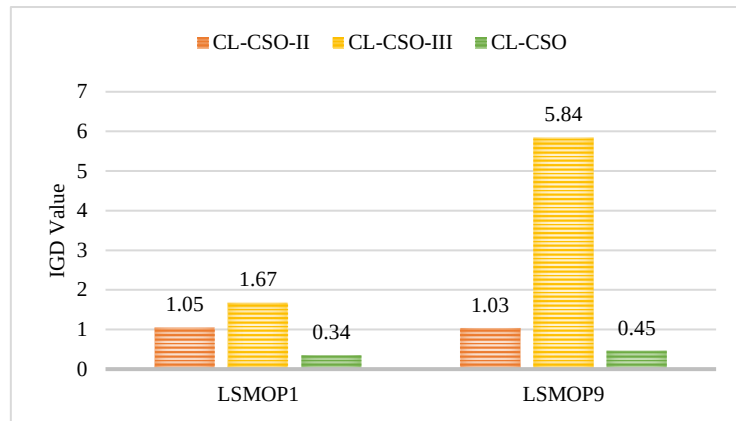


Fig. A. 6. IGD results achieved by CL-CSO-II, CLCSO-III and CL-CSO on 2-objective LSMOP1 with 2000 decision variables and 2-objective LSMOP9 with 2000 decision variables.

1.2 Tables:

Table A. 1: Search strategies and parameters settings of all the compared algorithms.

Algorithms	Search Strategies	Parameter settings
FDV	Proposed transformation +DE +PM	$Rate = 0.8, Acc = 0.4, optimizer = \text{"NSGA-II"}, type = 1, p_c = 1.0, p_m = 1/D, n_c = 20, n_m = 20$
LMOCSSO	Proposed CSO +PM	$p_m = 1/D, n_m = 20, a = 2$
LMEA	Proposed Clustering Variable Analysis +SBX +PM	$p_c = 1.0, p_m = 1/D, n_c = 20, n_m = 20, nSel = 2, nPer = 4, nCor = 6$
S-ECSO	Proposed CSO +Proposed Convex Sparse Operator	$\lambda_{initial} = 0.35, w = 0.7968, c_1 = 1.4962, c_2 = 1.4962$
LSMOEA/D	Proposed Clustering Variable Analysis + SBX +PM	$p_c = 1.0, p_m = 1/D, n_c = 20, n_m = 20, T = 0.1 \times N, nSel = 2, nPer = 4, K = 10$
CCSO	Proposed CSO +PM	$p_m = 1/D, n_m = 20$
NN-CSO	Neural Network Model +corresponding CSO	$K=5, E_{min} = 0.01, lr = 0.01, epochs = 20$
CL-CSO	Proposed CSO +PM	$T = 10, p = 0.9, a = 2$

Table A. 2: The IGD results of CL-CSO and its seven compared algorithms on solving LSMOP test problems with two objectives and 500-5000 decision variables.

Problem	M	D	FDV	LMEA	LMOC SO	S-ESCO	LSMOEA/D	CCSO	NN-CSO	CL-CSO
LSMOP1	2	500	2.7556e+0(3.26e-1)-	1.0833e+1(2.27e-1)-	1.4958e+0(3.68e-2)-	7.0711e-1(1.20e-16)-	9.8516e+00(7.93e-01)-	7.1650e-01(6.22e-02)-	7.1764e-1 (1.24e-1) -	3.8077e-1(2.13e-2)
LSMOP1	2	1000	3.2281e+0(4.04e-1)-	1.1312e+1(2.70e-1)-	1.5465e+0(8.60e-2)-	7.0711e-1(1.20e-16)-	1.0891e+01(2.67e-01)-	8.7893e-01(7.42e-02)-	6.4066e-1 (2.94e-1) -	3.4523e-1(6.64e-3)
LSMOP1	2	2000	3.9980e+0(1.85e-1)-	1.1411e+1(1.75e-1)-	1.5548e+0(4.63e-2)-	7.0711e-1(1.20e-16)-	1.1356e+01(2.39e-01)-	9.8836e-01(1.97e-02)-	5.7778e-1 (3.25e-1) -	3.4374e-1(6.26e-3)
LSMOP1	2	5000	4.8869e+0(1.74e-1)-	1.1583e+1(2.34e-1)-	1.5627e+0(3.13e-2)-	6.9812e-1(0.00e+0)-	1.1495e+01(1.80e-01)-	1.1680e+00(2.79e-02)-	5.4027e-1 (1.72e-1) -	3.3341e-1(9.40e-3)
LSMOP2	2	500	6.5232e-2(9.64e-4)-	7.3867e-2(2.07e-4)-	4.7644e-2(1.07e-3)-	6.6538e-2(0.00e+0)-	7.8419e-02(2.72e-03)-	4.3240e-02(8.82e-04)-	1.5533e-2 (7.43e-4) -	1.3912e-2(3.10e-4)
LSMOP2	2	1000	3.7007e-2(1.48e-4)-	4.0541e-2(1.47e-4)-	2.5942e-2(6.61e-4)-	3.8220e-2(7.49e-18)-	4.4743e-02(2.62e-03)-	2.5874e-02(6.07e-04)-	9.8540e-3 (2.26e-3) -	8.5338e-3(3.04e-4)
LSMOP2	2	2000	2.1078e-2(1.18e-4)-	2.3272e-2(6.77e-4)-	1.4351e-2(1.69e-4)-	2.2341e-2(0.00e+0)-	2.9288e-02(4.75e-03)-	1.4968e-02(3.30e-04)-	6.5216e-3 (6.42e-4) -	5.7528e-3(2.30e-4)
LSMOP2	2	5000	1.0919e-2(1.28e-4)-	1.3339e-2(5.54e-4)-	7.4813e-3(1.39e-4)-	7.0770e-3(9.37e-19)-	2.1733e-02(6.40e-03)-	7.9699e-03(3.46e-04)-	6.4300e-3 (7.38e-4) -	4.5639e-3(2.42e-4)
LSMOP3	2	500	1.9044e+1(1.34e+0)-	8.7174e+2(9.61e+2)-	1.7826e+1(5.11e+0)-	1.5721e+0(2.40e-16)+	3.6941e+01(4.18e+00)-	1.6518e+01(4.58e-01)-	2.0430e+0 (1.23e+0) +	5.1118e+0(8.12e-1)
LSMOP3	2	1000	2.0994e+1(1.19e+0)-	1.7704e+3(1.70e+3)-	1.6657e+1(2.89e+0)-	1.5774e+0(2.40e-16)+	3.8230e+01(3.75e+00)-	1.7471e+01(9.65e-01)-	2.3864e+0 (1.44e+0) =	3.7807e+0(7.97e-1)
LSMOP3	2	2000	2.3743e+1(7.92e-1)-	1.0345e+3(8.80e+2)-	1.3498e+1(5.01e+0)-	1.5804e+0(2.40e-16)+	3.9278e+01(1.25e+00)-	1.7911e+01(4.16e-01)-	2.1994e+0 (1.62e+0) +	3.5275e+0(7.10e-1)
LSMOP3	2	5000	2.7020e+1(6.47e-1)-	1.9441e+3(1.54e+3)-	1.1205e+1(3.38e+0)-	1.5765e+0(2.40e-16)+	4.0653e+01(8.39e-01)-	1.8723e+01(1.21e-01)-	1.9203e+0 (8.83e-1) +	3.5772e+0(2.86e-1)
LSMOP4	2	500	1.0481e-1(3.13e-3)-	1.3664e-1(3.85e-4)-	9.1732e-2(9.93e-4)-	1.2884e-1(3.00e-17)-	1.3649e-01(3.61e-03)-	8.8946e-02(6.59e-04)-	5.0758e-2 (2.65e-3) -	4.6413e-2(2.83e-3)
LSMOP4	2	1000	6.6225e-2(7.31e-4)-	7.8173e-2(3.59e-4)-	5.2797e-2(6.02e-4)-	7.4308e-2(1.50e-17)-	8.1991e-02(4.98e-03)-	5.2477e-02(5.95e-04)-	2.7888e-2 (1.50e-3) -	2.4698e-2(3.99e-4)
LSMOP4	2	2000	3.8441e-2(6.54e-4)-	4.4646e-2(4.25e-4)-	3.0176e-2(2.95e-4)-	4.1989e-2(7.49e-18)-	5.1775e-02(2.57e-03)-	3.0381e-02(4.66e-04)-	1.8411e-2 (1.81e-3) -	1.4325e-2(5.85e-4)
LSMOP4	2	5000	1.9098e-2(1.80e-4)-	2.2889e-2(4.32e-4)-	1.4783e-2(1.21e-4)-	1.2361e-2(1.87e-18)-	3.5459e-02(8.36e-03)-	1.5255e-02(2.77e-04)-	9.3943e-3 (6.09e-4) -	7.3855e-3(1.81e-4)
LSMOP5	2	500	5.8976e+0(1.94e+0)-	2.3219e+1(5.27e-1)-	3.2315e+0(2.03e-1)-	7.4209e-1(1.20e-16)=	2.3402e+01(6.58e-01)-	1.5285e+00(1.85e-01)-	1.5133e+0 (8.30e-1) -	7.4209e-1(1.20e-16)
LSMOP5	2	1000	6.4829e+0(1.41e+0)-	2.3737e+1(4.50e-1)-	3.2762e+0(1.45e-1)-	7.4209e-1(1.20e-16)=	2.4032e+01(7.60e-01)-	1.7302e+00(9.60e-02)-	1.6536e+0 (9.13e-1) =	7.4209e-1(1.20e-16)
LSMOP5	2	2000	7.4414e+0(2.13e+0)-	2.4128e+1(2.01e-1)-	3.2430e+0(7.28e-2)-	7.4209e-1(1.20e-16)=	2.4130e+01(3.65e-01)-	1.9757e+00(1.18e-01)-	1.1451e+0 (6.42e-1) =	7.4209e-1(1.20e-16)
LSMOP5	2	5000	9.7591e+0(1.67e+0)-	2.4433e+1(2.12e-1)-	3.2653e+0(1.04e-1)-	7.4209e-1(1.20e-16)=	2.4370e+01(3.51e-01)-	2.4064e+00(1.76e-01)-	1.2155e+0 (4.81e-1) -	7.4209e-1(1.20e-16)
LSMOP6	2	500	8.0740e-1(2.80e-3)-	3.5461e+3(2.58e+3)-	8.0728e-1(7.23e-3)-	7.0814e-1(1.20e-16)+	5.7178e-01(7.61e-02)+	7.3206e-01(9.66e-02)+	4.9883e-1 (2.02e-2) +	7.7331e-1(3.19e-3)
LSMOP6	2	1000	7.7413e-1(3.12e-4)-	2.7187e+3(2.59e+3)-	7.7439e-1(2.33e-3)-	6.8586e-1(1.20e-16)+	9.9963e-01(1.30e+00)-	6.1090e-01(1.55e-01)+	4.3973e-1 (1.14e-2) +	7.5544e-1(2.18e-3)
LSMOP6	2	2000	7.5738e-1(4.76e-13)-	4.0962e+3(2.18e+3)-	7.4755e-1(2.50e-2)+	6.7636e-1(1.20e-16)+	4.5906e-01(6.89e-02)+	5.4291e-01(1.47e-01)=	9.2869e-1 (8.69e-1) =	7.4789e-1(1.21e-3)
LSMOP6	2	5000	7.4774e-1(3.34e-13)-	2.6494e+3(2.31e+3)-	7.4774e-1(3.27e-13)-	1.7254e-1(3.00e-17)+	6.3853e-01(3.78e-01)=	6.9022e-01(1.32e-01)=	1.0196e+0 (7.52e-1) -	7.4435e-1(3.35e-4)
LSMOP7	2	500	8.4172e+2(1.16e+2)-	8.0776e+4(3.94e+3)-	8.0961e+2(1.30e+2)-	1.5118e+0(0.00e+0)=	7.7632e+04(5.13e+03)-	2.0971e+02(7.02e+01)-	1.5126e+0 (5.23e-4) =	1.5118e+0(2.00e-3)
LSMOP7	2	1000	1.2563e+3(1.72e+2)-	8.7506e+4(1.81e+3)-	8.8738e+2(1.40e+2)-	1.5189e+0(2.40e-16)-	8.3159e+04(3.52e+03)-	1.4026e+02(2.88e+01)-	1.5193e+0 (2.16e-4) -	1.5151e+0(1.02e-3)
LSMOP7	2	2000	2.0842e+3(4.40e+2)-	8.9829e+4(2.70e+3)-	9.3441e+2(7.70e+1)-	1.5224e+0(0.00e+0)-	8.6198e+04(3.00e+03)-	1.5776e+02(1.00e+02)-	1.5212e+0 (1.30e-3) -	1.5171e+0(7.01e-4)
LSMOP7	2	5000	3.4897e+3(1.98e+2)-	9.2604e+4(1.66e+3)-	1.0292e+3(1.05e+2)-	1.5173e+0(0.00e+0)=	9.0392e+04(1.91e+03)-	2.2699e+02(1.51e+02)-	1.5206e+0 (1.61e-3) -	1.5172e+0(7.06e-4)
LSMOP8	2	500	4.4066e+0(3.42e-1)-	1.9918e+1(3.99e-1)-	2.3245e+0(1.42e-1)-	7.4209e-1(1.20e-16)=	1.8934e+01(6.76e-01)-	1.1250e+00(1.03e-01)-	1.0448e+0 (2.69e-1) -	7.4209e-1(1.20e-16)
LSMOP8	2	1000	5.4678e+0(2.64e-1)-	2.0377e+1(3.21e-1)-	2.2645e+0(1.29e-1)-	7.4209e-1(1.20e-16)=	1.9855e+01(2.78e-01)-	1.2561e+00(4.30e-02)-	9.1094e-1 (1.77e-1) -	7.4209e-1(1.20e-16)
LSMOP8	2	2000	7.0737e+0(4.24e-1)-	2.0195e+1(3.76e-1)-	2.3436e+0(8.55e-2)-	7.4209e-1(1.20e-16)=	2.0329e+01(1.77e-01)-	1.4308e+00(7.28e-02)-	9.2829e-1 (1.54e-1) -	7.4209e-1(1.20e-16)
LSMOP8	2	5000	7.7684e+0(2.97e-1)-	2.0844e+1(2.37e-1)-	2.3850e+0(8.13e-2)-	7.4209e-1(1.20e-16)=	2.0817e+01(2.64e-01)-	1.7022e+00(3.52e-02)-	7.8350e-1 (4.04e-2) -	7.4209e-1(1.20e-16)
LSMOP9	2	500	1.2940e+0(2.76e-1)-	5.7192e+1(3.32e+0)-	1.1525e+0(1.34e-1)-	8.1004e-1(0.00e+0)-	3.4418e+01(3.22e+00)-	4.7438e-01(1.60e-02)=	8.1962e-1 (1.84e-2) -	5.3561e-1(9.16e-2)
LSMOP9	2	1000	1.5129e+0(2.10e-1)-	5.8805e+1(2.25e+0)-	1.7649e+0(4.14e-1)-	8.1004e-1(0.00e+0)-	4.0541e+01(3.81e+00)-	4.9522e-01(6.37e-03)-	8.3379e-1 (1.87e-2) -	4.8714e-1(1.26e-1)
LSMOP9	2	2000	1.9383e+0(1.92e-1)-	6.0675e+1(1.68e+0)-	2.3323e+0(5.08e-1)-	8.1004e-1(0.00e+0)-	4.8032e+01(2.61e+00)-	5.4251e-01(2.00e-02)=	7.8128e-1 (8.82e-2) -	4.5785e-1(1.78e-1)
LSMOP9	2	5000	3.9105e+0(6.55e-1)-	6.1156e+1(9.81e-1)-	5.1807e+0(1.67e+0)-	8.1004e-1(0.00e+0)-	5.1929e+01(3.45e+00)-	6.9323e-01(3.41e-02)-	7.6860e-1 (6.89e-2) -	5.1708e-1(2.57e-2)
+/-/=			0/36/0	0/36/0	1/35/0	8/18/10	2/33/1	2/30/4	5/26/5	

Table A. 3: The HV results of CL-CSO and its seven compared algorithms on solving LSMOP test problems with two objectives and 500-5000 decision variables.

Problem	M	D	FDV	LMEA	LMOCSO	S-ECSO	LSMOEA/D	CCSO	NN-CSO	CL-CSO
LSMOP1	2	500	0.0000e+0 (0.00e+0) -	0.0000e+0 (0.00e+0) -	0.0000e+0 (0.00e+0) -	9.0909e-2 (0.00e+0) -	0.0000e+0(0.00e+0)-	0.0000e+0(0.00e+0)-	9.6986e-3(2.57e-2)-	2.0287e-1 (1.25e-2)
LSMOP1	2	1000	0.0000e+0 (0.00e+0) -	0.0000e+0 (0.00e+0) -	0.0000e+0 (0.00e+0) -	9.0909e-2 (0.00e+0) -	0.0000e+0(0.00e+0)-	0.0000e+0(0.00e+0)-	9.2719e-2(1.16e-1)-	2.3043e-1 (4.43e-3)
LSMOP1	2	2000	0.0000e+0 (0.00e+0) -	0.0000e+0 (0.00e+0) -	0.0000e+0 (0.00e+0) -	9.0909e-2 (0.00e+0) -	0.0000e+0(0.00e+0)-	0.0000e+0(0.00e+0)-	1.3649e-1(9.96e-2)-	2.3196e-1 (4.80e-3)
LSMOP1	2	5000	0.0000e+0 (0.00e+0) -	0.0000e+0 (0.00e+0) -	0.0000e+0 (0.00e+0) -	9.2668e-2 (1.50e-17) -	0.0000e+0(0.00e+0)-	0.0000e+0(0.00e+0)-	1.0232e-1(7.60e-2)-	2.3959e-1 (6.16e-3)
LSMOP2	2	500	4.9990e-1 (1.21e-3) -	4.8712e-1 (7.74e-4) -	5.2240e-1 (1.35e-3) -	5.0173e-1 (0.00e+0) -	4.7573e-1(5.46e-3)-	4.8712e-1(3.25e-3)-	5.6501e-1(9.24e-04)-	5.6634e-1 (4.35e-4)
LSMOP2	2	1000	5.3565e-1 (2.83e-4) -	5.2887e-1 (3.77e-4) -	5.5074e-1 (8.82e-4) -	5.3560e-1 (0.00e+0) -	5.1883e-1(5.73e-3)-	4.9983e-1(5.80e-3)-	5.7218e-1(2.95e-3)=	5.7317e-1 (6.07e-4)
LSMOP2	2	2000	5.5676e-1 (3.60e-4) -	5.5185e-1 (1.19e-3) -	5.6601e-1 (2.33e-4) -	5.5523e-1 (0.00e+0) -	5.3962e-1(7.39e-3)-	5.1866e-1(1.02e-2)-	5.7665e-1(9.60e-04)-	5.7735e-1 (3.59e-4)
LSMOP2	2	5000	5.7017e-1 (1.73e-4) -	5.6544e-1 (1.13e-3) -	5.7521e-1 (1.97e-4) -	5.7588e-1 (0.00e+0) -	5.5118e-1(9.07e-3)-	5.3163e-1(5.42e-3)-	5.7687e-1(1.09e-3)-	5.7949e-1 (1.59e-4)
LSMOP3	2	500	0.0000e+0 (0.00e+0) =	0.0000e+0 (0.00e+0) =	0.0000e+0 (0.00e+0) =	0.0000e+0 (0.00e+0) =	0.0000e+0(0.00e+0)=	0.0000e+0(0.00e+0)=	0.0000e+0(0.00e+0)=	0.0000e+0 (0.00e+0)
LSMOP3	2	1000	0.0000e+0 (0.00e+0) =	0.0000e+0 (0.00e+0) =	0.0000e+0 (0.00e+0) =	0.0000e+0 (0.00e+0) =	0.0000e+0(0.00e+0)=	0.0000e+0(0.00e+0)=	0.0000e+0(0.00e+0)=	0.0000e+0 (0.00e+0)
LSMOP3	2	2000	0.0000e+0 (0.00e+0) =	0.0000e+0 (0.00e+0) =	0.0000e+0 (0.00e+0) =	0.0000e+0 (0.00e+0) =	0.0000e+0(0.00e+0)=	0.0000e+0(0.00e+0)=	0.0000e+0(0.00e+0)=	0.0000e+0 (0.00e+0)
LSMOP3	2	5000	0.0000e+0 (0.00e+0) =	0.0000e+0 (0.00e+0) =	0.0000e+0 (0.00e+0) =	0.0000e+0 (0.00e+0) =	0.0000e+0(0.00e+0)=	0.0000e+0(0.00e+0)=	0.0000e+0(0.00e+0)=	0.0000e+0 (0.00e+0)
LSMOP4	2	500	4.5187e-1 (3.88e-3) -	4.1512e-1 (9.58e-4) -	4.6983e-1 (1.08e-3) -	4.2429e-1 (6.00e-17) -	4.1194e-1(7.52e-3)-	4.2508e-1(9.75e-3)-	5.2178e-1(3.18e-3)=	5.2368e-1 (3.42e-3)
LSMOP4	2	1000	4.9849e-1 (1.01e-3) -	4.8178e-1 (1.05e-3) -	5.1584e-1 (7.21e-4) -	4.8799e-1 (0.00e+0) -	4.7220e-1(8.05e-3)-	4.7876e-1(4.59e-3)-	5.5131e-1(1.95e-3)=	5.5208e-1 (4.43e-4)
LSMOP4	2	2000	5.3362e-1 (8.55e-4) -	5.2314e-1 (1.81e-3) -	5.4465e-1 (3.70e-4) -	5.2839e-1 (1.20e-16) -	5.0606e-1(3.26e-3)-	4.9832e-1(9.28e-3)-	5.6358e-1(2.43e-3)=	5.6576e-1 (7.08e-4)
LSMOP4	2	5000	5.5929e-1 (3.36e-4) -	5.5205e-1 (1.44e-3) -	5.6536e-1 (1.59e-4) -	5.6875e-1 (0.00e+0) -	5.2979e-1(8.75e-3)-	5.1914e-1(7.68e-3)-	5.7270e-1(8.20e-04)-	5.7501e-1 (3.07e-4)
LSMOP5	2	500	0.0000e+0 (0.00e+0) -	0.0000e+0 (0.00e+0) -	0.0000e+0 (0.00e+0) -	9.0909e-2 (0.00e+0) =	0.0000e+0(0.00e+0)-	0.0000e+0(0.00e+0)-	1.2987e-2(3.44e-2)-	9.0909e-2 (0.00e+0)
LSMOP5	2	1000	0.0000e+0 (0.00e+0) -	0.0000e+0 (0.00e+0) -	0.0000e+0 (0.00e+0) -	9.0909e-2 (0.00e+0) =	0.0000e+0(0.00e+0)-	0.0000e+0(0.00e+0)-	2.5974e-2(4.44e-2)-	9.0909e-2 (0.00e+0)
LSMOP5	2	2000	0.0000e+0 (0.00e+0) -	0.0000e+0 (0.00e+0) -	0.0000e+0 (0.00e+0) -	9.0909e-2 (0.00e+0) =	0.0000e+0(0.00e+0)-	0.0000e+0(0.00e+0)-	2.8561e-2(4.31e-2)-	9.0909e-2 (0.00e+0)
LSMOP5	2	5000	0.0000e+0 (0.00e+0) -	0.0000e+0 (0.00e+0) -	0.0000e+0 (0.00e+0) -	9.0909e-2 (0.00e+0) =	0.0000e+0(0.00e+0)-	0.0000e+0(0.00e+0)-	1.2987e-2(3.44e-2)-	9.0909e-2 (0.00e+0)
LSMOP6	2	500	0.0000e+0 (0.00e+0) -	0.0000e+0 (0.00e+0) -	0.0000e+0 (0.00e+0) -	0.0000e+0 (0.00e+0) -	0.0000e+0(0.00e+0)-	9.7828e-04(1.68e-3)-	2.1349e-3(5.65e-3)-	3.0815e-2 (5.40e-3)
LSMOP6	2	1000	2.9421e-2 (5.22e-4) -	0.0000e+0 (0.00e+0) -	2.8988e-2 (3.89e-3) -	2.6889e-2 (0.00e+0) -	1.9839e-2(9.93e-3)-	4.9343e-2(2.07e-3)-	3.2782e-2(7.58e-3)-	6.2726e-2 (4.17e-3)
LSMOP6	2	2000	5.9011e-2 (3.00e-14) -	0.0000e+0 (0.00e+0) -	5.9023e-2 (2.08e-5) -	5.9011e-2 (1.50e-17) -	5.8712e-2(7.64e-04)-	7.1061e-2(6.70e-04)-	4.3729e-2(3.00e-2)-	7.7872e-2 (2.56e-3)
LSMOP6	2	5000	7.8177e-2 (2.71e-14) -	0.0000e+0 (0.00e+0) -	7.8177e-2 (2.65e-14) -	1.7253e-1 (0.00e+0) ‡	6.5215e-2(2.91e-2)-	8.3072e-2(2.12e-04)-	6.7587e-2(2.98e-2)-	8.5598e-2 (7.58e-4)
LSMOP7	2	500	0.0000e+0 (0.00e+0) =	0.0000e+0 (0.00e+0) =	0.0000e+0 (0.00e+0) =	0.0000e+0 (0.00e+0) =	0.0000e+0(0.00e+0)=	0.0000e+0(0.00e+0)=	0.0000e+0(0.00e+0)=	0.0000e+0 (0.00e+0)
LSMOP7	2	1000	0.0000e+0 (0.00e+0) =	0.0000e+0 (0.00e+0) =	0.0000e+0 (0.00e+0) =	0.0000e+0 (0.00e+0) =	0.0000e+0(0.00e+0)=	0.0000e+0(0.00e+0)=	0.0000e+0(0.00e+0)=	0.0000e+0 (0.00e+0)
LSMOP7	2	2000	0.0000e+0 (0.00e+0) =	0.0000e+0 (0.00e+0) =	0.0000e+0 (0.00e+0) =	0.0000e+0 (0.00e+0) =	0.0000e+0(0.00e+0)=	0.0000e+0(0.00e+0)=	0.0000e+0(0.00e+0)=	0.0000e+0 (0.00e+0)
LSMOP7	2	5000	0.0000e+0 (0.00e+0) =	0.0000e+0 (0.00e+0) =	0.0000e+0 (0.00e+0) =	0.0000e+0 (0.00e+0) =	0.0000e+0(0.00e+0)=	0.0000e+0(0.00e+0)=	0.0000e+0(0.00e+0)=	0.0000e+0 (0.00e+0)
LSMOP8	2	500	0.0000e+0 (0.00e+0) -	0.0000e+0 (0.00e+0) -	0.0000e+0 (0.00e+0) -	9.0909e-2 (0.00e+0) =	0.0000e+0(0.00e+0)-	0.0000e+0(0.00e+0)-	2.4921e-2(4.26e-2)-	9.0909e-2 (0.00e+0)
LSMOP8	2	1000	0.0000e+0 (0.00e+0) -	0.0000e+0 (0.00e+0) -	0.0000e+0 (0.00e+0) -	9.0909e-2 (0.00e+0) =	0.0000e+0(0.00e+0)-	0.0000e+0(0.00e+0)-	2.5974e-2(4.44e-2)-	9.0909e-2 (0.00e+0)
LSMOP8	2	2000	0.0000e+0 (0.00e+0) -	0.0000e+0 (0.00e+0) -	0.0000e+0 (0.00e+0) -	9.0909e-2 (0.00e+0) =	0.0000e+0(0.00e+0)-	0.0000e+0(0.00e+0)-	0.0000e+0(0.00e+0)-	9.0909e-2 (0.00e+0)
LSMOP8	2	5000	0.0000e+0 (0.00e+0) -	0.0000e+0 (0.00e+0) -	0.0000e+0 (0.00e+0) -	9.0909e-2 (0.00e+0) =	0.0000e+0(0.00e+0)-	0.0000e+0(0.00e+0)-	2.0927e-2(3.36e-2)-	9.0909e-2 (0.00e+0)
LSMOP9	2	500	6.3944e-3 (1.40e-2) -	0.0000e+0 (0.00e+0) -	3.8570e-3 (5.19e-3) -	9.0909e-2 (0.00e+0) =	0.0000e+0(0.00e+0)=	0.0000e+0(0.00e+0)=	0.0000e+0(0.00e+0)=	1.1612e-1 (4.05e-2)
LSMOP9	2	1000	0.0000e+0 (0.00e+0) -	0.0000e+0 (0.00e+0) -	9.8617e-4 (2.61e-3) -	9.0909e-2 (0.00e+0) -	0.0000e+0(0.00e+0)=	0.0000e+0(0.00e+0)=	0.0000e+0(0.00e+0)=	1.5261e-1 (1.35e-2)
LSMOP9	2	2000	0.0000e+0 (0.00e+0) -	0.0000e+0 (0.00e+0) -	0.0000e+0 (0.00e+0) -	9.0909e-2 (0.00e+0) -	0.0000e+0(0.00e+0)=	0.0000e+0(0.00e+0)=	0.0000e+0(0.00e+0)=	1.6044e-1 (2.59e-2)
LSMOP9	2	5000	0.0000e+0 (0.00e+0) -	0.0000e+0 (0.00e+0) -	0.0000e+0 (0.00e+0) -	9.0909e-2 (0.00e+0) -	0.0000e+0(0.00e+0)=	0.0000e+0(0.00e+0)=	0.0000e+0(0.00e+0)=	1.5331e-1 (7.20e-3)
+/-/=			0/28/8	0/28/8	0/28/8	1/18/17	0/24/12	0/24/12	0/20/16	

Table A. 4: The IGD results of CL-CSO and its seven compared algorithms on solving LSMOP test problems with three objectives and 500-5000 decision variables.

Problem	M	D	FDV	LMEA	LMOCSO	S-ECSO	LSMOEA/D	CCSO	NN-CSO	CL-CSO
LSMOP1	3	500	3.9008e+0 (1.91e-1) -	1.0938e+1 (9.74e-1) -	1.6044e+0 (1.09e-1) -	6.0027e-1 (1.20e-16) -	1.0078e+1(6.55e-1)-	1.1834e+0(9.47e-2)-	6.9294e-1 (1.49e-1) -	4.7282e-1 (3.12e-2)
LSMOP1	3	1000	5.9723e+0 (8.50e-1) -	1.0049e+1 (4.43e+0) -	1.5830e+0 (1.09e-1) -	4.1580e-1 (0.00e+0) =	1.0697e+1(3.18e-1)-	1.2044e+0(2.60e-2)-	6.9418e-1 (3.56e-2) -	4.1979e-1 (2.08e-2)
LSMOP1	3	2000	7.4952e+0 (4.58e-1) -	1.1727e+1 (2.56e-1) -	1.5248e+0 (6.61e-2) -	4.2775e-1 (0.00e+0) -	1.0951e+1(3.23e-1)-	1.2393e+0(5.63e-2)-	5.8190e-1 (1.24e-1) -	3.7815e-1 (2.09e-2)
LSMOP1	3	5000	8.9938e+0(3.67e-1)-	1.1720e+1(2.85e-1)-	1.5858e+0(2.66e-2)-	3.7608e-1(6.00e-17)-	1.1278e+1(2.40e-1)-	1.3313e+0(7.48e-2)-	8.4205e-1(1.58e-1)-	3.3341e-1(9.40e-3)
LSMOP2	3	500	8.8726e-2 (5.10e-3) -	8.0274e-2 (2.35e-3) -	6.1731e-2 (5.03e-4) -	6.9347e-2 (1.50e-17) -	1.1416e-1(1.02e-2)-	8.8534e-2(2.53e-3)-	5.7052e-2 (1.13e-3) -	5.0400e-2 (1.43e-3)
LSMOP2	3	1000	6.8273e-2 (2.98e-3) -	6.4157e-2 (2.81e-3) -	5.0224e-2 (3.81e-4) -	5.2796e-2 (7.49e-18) -	1.0598e-1(1.16e-2)-	8.1739e-2(1.62e-3)-	4.7517e-2 (2.40e-4) -	4.4292e-2 (8.38e-4)
LSMOP2	3	2000	6.2754e-2 (3.24e-3) -	6.2064e-2 (5.16e-3) -	4.5313e-2 (1.03e-4) -	4.5945e-2 (7.49e-18) -	9.5647e-2(2.04e-2)-	7.7264e-2(2.44e-3)-	4.3749e-2 (2.56e-4) -	4.2292e-2 (1.51e-3)
LSMOP2	3	5000	6.0988e-2(2.92e-3)-	5.7487e-2(4.30e-3)-	4.2782e-2(3.14e-05)-	4.2663e-2(0.00e+0)-	1.0242e-1(1.11e-2)-	7.5282e-2(2.31e-3)-	4.1930e-2(4.61e-05)-	4.5639e-3(2.42e-4)
LSMOP3	3	500	1.3862e+1 (2.44e+0) -	3.3915e+2 (3.92e+2) -	1.4075e+1 (4.38e+0) -	8.6072e-1 (1.20e-16) -	4.3326e+1(4.04e+1)-	8.4256e+0(6.61e-1)-	3.9293e+0 (3.01e+0) -	8.5984e-1 (1.41e-3)
LSMOP3	3	1000	1.6823e+1 (3.34e+0) -	5.8840e+2 (4.96e+2) -	1.2630e+1 (8.90e-1) -	8.6072e-1 (1.20e-16) -	1.9597e+1(6.26e-1)-	8.9423e+0(7.36e-1)-	8.1325e-1 (3.94e-2) +	8.6069e-1 (1.20e-16)
LSMOP3	3	2000	1.9100e+1 (0.00e+0) -	2.2894e+2 (3.07e-14) -	1.2943e+1 (1.92e-15) -	8.6072e-1 (1.20e-16) -	2.0195e+1(4.69e-1)-	9.3273e+0(4.06e-1)-	9.7014e-1 (1.13e-1) -	8.5878e-1 (1.20e-16)
LSMOP3	3	5000	2.5117e+1(3.84e-15)-	4.5948e+2(0.00e+0)-	1.3448e+1(0.00e+0)-	8.6072e-1(1.20e-16)+	2.0218e+1(5.27e-1)-	9.7234e+0(7.18e-1)-	8.4505e-1(1.30e-1)+	3.5772e+0(2.86e-1)
LSMOP4	3	500	2.0034e-1 (3.00e-17) -	2.2664e-1 (0.00e+0) -	1.6034e-1 (0.00e+0) -	1.5777e-1 (0.00e+0) -	2.2190e-1(1.78e-2)-	1.7331e-1(4.01e-3)-	1.2102e-1 (3.09e-3) -	1.0915e-1 (1.50e-17)
LSMOP4	3	1000	1.3020e-1 (0.00e+0) -	1.3249e-1 (0.00e+0) -	9.9126e-2 (1.50e-17) -	1.0303e-1 (1.50e-17) -	1.5067e-1(2.29e-2)-	1.2365e-1(1.79e-3)-	7.5018e-2 (1.20e-3) -	6.5371e-2 (0.00e+0)
LSMOP4	3	2000	9.2858e-2 (0.00e+0) -	8.6372e-2 (1.50e-17) -	6.6483e-2 (0.00e+0) -	6.8772e-2 (1.50e-17) -	1.1943e-1(1.83e-2)-	9.6960e-2(1.46e-3)-	5.4678e-2 (6.94e-4) -	5.0959e-2 (0.00e+0)
LSMOP4	3	5000	6.9228e-2(0.00e+0)-	7.4303e-2(1.50e-17)-	4.8864e-2(7.49e-18)-	4.9338e-2(7.49e-18)-	1.0893e-1(2.24e-2)-	8.0369e-2(4.55e-3)-	4.4943e-2(2.61e-4)-	7.3855e-3(1.81e-4)
LSMOP5	3	500	1.2330e+1 (1.92e-15) -	2.0694e+1 (3.84e-15) -	3.2483e+0 (0.00e+0) -	5.8276e-1 (0.00e+0) -	1.7851e+1(9.60e-1)-	2.1585e+0(1.10e-1)-	6.1330e-1 (1.25e-1) -	4.6107e-1 (6.00e-17)
LSMOP5	3	1000	1.2660e+1 (0.00e+0) -	2.0358e+1 (0.00e+0) -	3.6798e+0 (4.80e-16) -	5.4203e-1 (0.00e+0) +	1.9017e+1(6.64e-1)-	2.2999e+0(1.15e-1)-	6.5239e-1 (2.93e-1) +	2.1454e+0 (0.00e+0)
LSMOP5	3	2000	1.7984e+1 (0.00e+0) -	2.0226e+1 (0.00e+0) -	3.2008e+0 (0.00e+0) -	5.3318e-1 (1.20e-16) +	1.9828e+1(6.01e-1)-	2.3734e+0(1.17e-1)-	8.8895e-1 (5.40e-1) +	2.2438e+0 (0.00e+0)
LSMOP5	3	5000	2.0692e+1(0.00e+0)-	2.0698e+1(3.84e-15)-	3.1318e+0(4.80e-16)-	5.3413e-1(1.20e-16)+	1.9859e+1(1.37e-1)-	2.5168e+0(7.53e-2)-	7.1214e-1(3.13e-1)=	7.4209e-1(0.00e+0)
LSMOP6	3	500	1.6083e+3 (2.46e-13) -	3.9025e+4 (0.00e+0) -	4.1591e+2 (0.00e+0) -	1.2929e+0 (2.40e-16) -	1.7375e+4(4.23e+3)-	3.2946e+1(2.74e+1)-	1.4796e+0 (1.69e-1) -	1.2915e+0 (2.40e-16)
LSMOP6	3	1000	2.7941e+3 (0.00e+0) -	5.1351e+4 (0.00e+0) -	2.8428e+2 (0.00e+0) -	1.6801e+0 (2.40e-16) -	2.2105e+4(2.44e+3)-	3.5339e+1(3.48e+1)-	1.3500e+0 (1.12e-1) =	1.3090e+0 (2.40e-16)
LSMOP6	3	2000	1.2306e+4 (0.00e+0) -	3.7846e+4 (0.00e+0) -	2.6417e+2 (6.14e-14) -	1.3231e+0 (0.00e+0) -	2.1990e+4(2.95e+3)-	5.3249e+1(3.78e+1)-	1.4809e+0 (1.04e-1) -	1.3192e+0 (2.40e-16)
LSMOP6	3	5000	3.5753e+4(7.86e-12)-	3.4728e+4(0.00e+0)-	3.4663e+2(0.00e+0)-	1.3253e+0(0.00e+0)-	2.2472e+4(1.94e+3)-	5.6353e+1(3.91e+1)-	1.3834e+0(1.04e-1)-	7.4435e-1(3.35e-4)
LSMOP7	3	500	1.2781e+0 (0.00e+0) -	9.1861e+2 (0.00e+0) -	1.0624e+0 (2.40e-16) -	8.7312e-1 (1.20e-16) -	1.1338e+0(3.74e-2)-	1.1127e+0(6.81e-2)-	8.8566e-1 (4.14e-2) -	8.0230e-1 (1.20e-16)
LSMOP7	3	1000	1.1070e+0 (2.40e-16) -	7.9014e+2 (0.00e+0) -	9.6363e-1 (0.00e+0) +	7.7530e-1 (1.20e-16) +	1.0038e+0(1.38e-1)=	9.3105e-1(9.75e-2)=	9.3940e-1 (9.69e-2) =	1.0078e+0 (0.00e+0)
LSMOP7	3	2000	1.0206e+0 (0.00e+0) -	1.3731e+3 (2.46e-13) -	9.8056e-1 (0.00e+0) -	7.8423e-1 (1.20e-16) +	7.9278e-1(1.11e-1)+	8.4365e-1(3.29e-2)+	8.3630e-1 (1.72e-1) +	9.7503e-1 (0.00e+0)
LSMOP7	3	5000	9.7266e-1(1.20e-16)+	1.5591e+3(2.46e-13)-	9.5572e-1(1.20e-16)+	7.4695e-1(1.20e-16)+	8.6115e-1(1.37e-1)+	8.9322e-1(6.28e-2)+	6.5391e-1(1.69e-1)+	1.5172e+0(7.06e-4)
LSMOP8	3	500	9.6746e-1 (0.00e+0) -	6.8232e-1 (1.20e-16) +	7.6256e-1 (0.00e+0) +	3.2348e-1 (0.00e+0) +	6.8576e-1(5.49e-2)+	6.3889e-1(6.37e-2)+	8.7354e-1 (7.43e-2) =	8.6099e-1 (0.00e+0)
LSMOP8	3	1000	9.5826e-1 (0.00e+0) -	6.1358e-1 (1.20e-16) +	9.5319e-1 (1.20e-16) +	1.1575e-1 (0.00e+0) +	6.3896e-1(3.48e-2)+	7.5221e-1(1.02e-1)+	6.1471e-1 (4.08e-1) +	9.5385e-1 (0.00e+0)
LSMOP8	3	2000	9.5363e-1 (1.20e-16) -	6.3818e-1 (0.00e+0) +	9.4368e-1 (0.00e+0) +	9.5922e-2 (0.00e+0) +	6.0356e-1(2.08e-2)+	8.3981e-1(1.10e-1)+	1.1622e-1 (1.74e-2) +	9.5039e-1 (0.00e+0)
LSMOP8	3	5000	9.5132e-1(1.20e-16)-	6.1878e-1(1.20e-16)+	9.4722e-1(2.40e-16)-	9.1598e-2(0.00e+0)+	6.5753e-1(7.85e-2)+	8.8918e-1(7.57e-2)-	1.1225e-1(2.07e-2)+	7.4209e-1(0.00e+0)
LSMOP9	3	500	6.3790e+0 (9.59e-16) -	1.3726e+2 (0.00e+0) -	7.4751e+0 (9.59e-16) -	1.5379e+0 (2.40e-16) -	8.0065e+1(7.49e+0)-	1.6999e+0(2.32e-1)-	8.0827e-1 (2.47e-1) =	1.1415e+0 (0.00e+0)
LSMOP9	3	1000	7.3806e+0 (0.00e+0) +	1.4478e+2 (3.07e-14) -	6.6805e+1 (0.00e+0) -	5.9286e-1 (1.20e-16) +	8.2171e+1(6.35e+0)-	1.7075e+0(1.21e-1)+	9.4675e-1 (3.54e-1) +	1.0910e+1 (1.92e-15)
LSMOP9	3	2000	1.5787e+1 (0.00e+0) +	1.4667e+2 (0.00e+0) -	1.9304e+2 (3.07e-14) -	6.9983e-1 (0.00e+0) +	8.6464e+1(4.64e+0)-	2.2268e+0(1.87e-1)+	7.7742e-1 (3.65e-1) +	2.8230e+1 (3.84e-15)
LSMOP9	3	5000	2.5152e+1(3.84e-15)-	1.5306e+2(3.07e-14)-	1.4795e+2(0.00e+0)-	1.5379e+0(2.40e-16)-	8.7253e+1(6.55e+0)-	3.1998e+0(1.35e-1)-	7.5906e-1(2.20e-1)-	5.1708e-1(2.57e-2)
+/-/=			3/33/0	4/32/0	5/31/0	13/22/1	6/29/1	7/28/1	11/20/5	

Table A. 5: The IGD results of CL-CSO and its seven compared algorithms on solving IMF test problems with two objectives and 500-5000 decision variables.

Problem	M	D	FDV	LMEA	LMOC SO	S-ESCO	LSMOEA/D	CCSO	NN-CSO	CL-CSO
IMF1	2	500	5.5970e-1(3.44e-2)	1.5657e+1(6.45e-1)	5.5752e+0(1.07e-1)	6.0132e-1(0.00e+0)	1.2638e+01(9.02e-01)	1.1358e+00(8.02e-02)	4.3845e-1 (8.75e-2) -	3.8031e-1(2.95e-2)
IMF1	2	1000	7.1922e-1(4.92e-2)	1.5593e+1(7.36e-1)	6.2005e+0(2.45e-1)	6.1507e-1(0.00e+0)	1.3843e+01(5.59e-01)	1.2740e+00(5.28e-02)	4.9834e-1 (1.04e-1) -	3.4168e-1(5.69e-2)
IMF1	2	2000	8.2546e-1(3.08e-2)	1.6445e+1(5.22e-1)	6.6671e+0(2.06e-1)	5.9171e-1(0.00e+0)	1.4438e+01(4.44e-01)	1.2985e+00(5.31e-02)	4.6747e-1 (2.38e-2) -	3.5932e-1(3.63e-2)
IMF1	2	5000	1.2072e+0(4.00e-2)	1.6561e+1(2.77e-1)	7.1473e+0(6.31e-2)	5.8822e-1(1.20e-16)	1.5520e+01(4.71e-01)	1.3467e+00(4.36e-02)	5.8349e-1 (5.03e-2) -	3.6090e-1(2.36e-2)
IMF2	2	500	9.8670e-1(8.94e-2)	1.9115e+1(8.40e-1)	8.0882e+0(1.46e-1)	6.0949e-1(1.20e-16)	1.6684e+01(1.14e+00)	1.9510e+00(6.71e-02)	6.3720e-1 (5.02e-2) =	6.0946e-1(2.54e-4)
IMF2	2	1000	1.1424e+0(6.22e-2)	1.9341e+1(8.10e-1)	8.5476e+0(3.06e-1)	6.0949e-1(1.20e-16) +	1.7611e+01(7.73e-01)	2.1204e+00(1.59e-01)	6.4447e-1 (5.08e-2) =	6.0949e-1(1.64e-8)
IMF2	2	2000	1.5926e+0(1.23e-1)	2.0441e+1(4.56e-1)	9.4274e+0(1.57e-1)	6.0949e-1(1.20e-16) =	1.9024e+01(5.73e-01)	2.2131e+00(4.86e-02)	7.8467e-1 (1.90e-1) =	6.0949e-1(1.81e-16)
IMF2	2	5000	2.2246e+0(1.11e-1)	2.0624e+1(5.16e-1)	9.9694e+0(9.17e-2)	6.0949e-1(1.20e-16) =	1.9418e+01(5.51e-01)	2.3230e+00(6.34e-02)	6.7302e-1 (8.49e-2) =	6.0949e-1(1.76e-16)
IMF3	2	500	1.1513e+0(2.18e-1)	1.9093e+1(1.03e+0)	8.1546e+0(1.71e-1)	2.2086e-1(0.00e+0)	1.6962e+01(9.55e-01)	1.9186e+00(5.08e-02)	6.9717e-2 (4.27e-2) -	1.0564e-2(2.19e-3)
IMF3	2	1000	1.4379e+0(1.95e-1)	2.0306e+1(9.69e-1)	8.8785e+0(2.05e-1)	1.6973e-1(0.00e+0)	1.8778e+01(5.53e-01)	1.9867e+00(7.91e-02)	1.6189e-1 (1.32e-1) -	8.8849e-1(3.18e-3)
IMF3	2	2000	1.8847e+0(1.55e-1)	2.0119e+1(6.73e-1)	9.1350e+0(3.99e-1)	2.3170e-1(3.00e-17)	1.9387e+01(4.30e-01)	1.9895e+00(1.21e-01)	8.7162e-2 (5.83e-2) -	8.4593e-1(3.62e-4)
IMF3	2	5000	2.5498e+0(1.69e-1)	2.0962e+1(3.62e-1)	9.3484e+0(1.84e-1)	5.2329e-2(0.00e+0)	1.9799e+01(2.53e-01)	2.2821e+00(7.80e-02)	6.6610e-2 (6.05e-2) -	8.7152e-1(9.58e-4)
IMF4	3	500	9.2068e+1(7.10e+0)	1.0587e+3(3.44e+1)	5.7504e+2(5.51e+1)	5.9890e-1(0.00e+0)	8.9571e+02(3.05e+01)	3.3803e+02(1.01e+02)	3.5431e+2 (3.32e+2) -	5.8523e-1(3.44e-4)
IMF4	3	1000	2.1004e+2(1.60e+1)	2.2005e+3(6.18e+1)	1.1936e+3(8.52e+1)	5.5068e-1(0.00e+0) +	1.8432e+03(4.98e+01)	6.6226e+02(2.73e+02)	1.1121e+3 (4.91e+2) -	5.5757e-1(3.25e-2)
IMF4	3	2000	5.2007e+2(2.79e+1)	4.4076e+3(8.06e+1)	2.3014e+3(1.17e+2)	5.7880e-1(0.00e+0)	3.8700e+03(1.05e+02)	1.0601e+03(5.35e+02)	2.2195e+3 (9.86e+2) -	5.4523e-1(3.94e-4)
IMF4	3	5000	2.1504e+3(1.15e+3)	1.1390e+4(1.25e+2)	5.9335e+3(1.93e+2)	5.5433e-1(0.00e+0)	1.0121e+04(2.10e+02)	1.4624e+03(1.13e+03)	4.2785e+3 (2.40e+3) -	5.4466e-1(3.28e-4)
IMF5	2	500	2.1541e-1(9.64e-3)	3.4009e-1(7.48e-3)	2.0104e-1(2.18e-3)	5.8591e-1(0.00e+0)	3.1520e-01(9.49e-03)	2.8673e-01(1.40e-02)	5.8041e-2 (4.60e-3) -	4.4784e-2(1.90e-3)
IMF5	2	1000	2.4006e-1(1.14e-2)	3.4318e-1(3.18e-3)	1.9884e-1(1.65e-3)	6.1505e-1(1.20e-16)	3.2885e-01(9.18e-03)	3.1187e-01(3.16e-02)	5.3515e-2 (2.39e-3) -	4.6515e-2(1.49e-3)
IMF5	2	2000	2.7079e-1(8.06e-3)	3.4762e-1(2.37e-3)	2.0056e-1(3.99e-3)	5.9155e-1(0.00e+0)	3.3988e-01(5.57e-03)	3.3505e-01(1.28e-02)	5.1172e-2 (2.70e-3) -	4.6305e-2(1.01e-3)
IMF5	2	5000	2.8486e-1(6.54e-3)	3.5429e-1(1.60e-3)	1.9939e-1(1.15e-3)	6.1739e-1(1.20e-16)	3.4573e-01(7.68e-03)	3.1903e-01(3.90e-02)	4.8505e-2 (1.98e-3) -	4.6560e-2(4.31e-4)
IMF6	2	500	3.7782e-1(1.54e-2)	6.0382e-1(2.19e-2)	2.9101e-1(5.18e-3)	6.0949e-1(1.20e-16)	5.4028e-01(1.39e-02)	3.8701e-01(4.22e-02)	9.4466e-2 (6.63e-3) =	9.1588e-2(1.87e-3)
IMF6	2	1000	4.2077e-1(8.64e-3)	6.0807e-1(1.07e-2)	2.8607e-1(1.95e-3)	6.0949e-1(1.20e-16)	5.7105e-01(1.91e-02)	3.8611e-01(4.11e-02)	8.4711e-2(5.41e-3) +	9.1462e-2(1.54e-3)
IMF6	2	2000	4.6773e-1(7.82e-3)	6.2208e-1(8.41e-3)	2.8628e-1(2.08e-3)	6.0949e-1(1.20e-16)	6.0204e-01(8.79e-03)	4.6220e-01(1.94e-02)	7.8863e-2(4.78e-3) +	9.1459e-2(9.31e-4)
IMF6	2	5000	5.1459e-1(1.59e-2)	6.3098e-1(5.21e-3)	2.8662e-1(5.93e-4)	6.0949e-1(1.20e-16)	6.1861e-01(1.23e-02)	4.5225e-01(2.14e-02)	9.4161e-2(2.04e-3) =	1.6830e-1(2.04e-1)
IMF7	2	500	3.2172e-1(8.55e-3)	5.2187e-1(1.24e-2)	3.0499e-1(3.46e-3)	1.4213e-1(0.00e+0)	4.6701e-01(1.24e-02)	6.9082e-01(2.74e-03)	8.2363e-2 (4.09e-2) -	2.0732e-2(7.77e-3)
IMF7	2	1000	3.4872e-1(9.71e-3)	5.2447e-1(4.79e-3)	3.0632e-1(4.43e-3)	1.6972e-1(3.00e-17)	5.0381e-01(1.70e-02)	6.8842e-01(3.68e-03)	4.9162e-2 (2.09e-2) -	2.1142e-2(7.62e-3)
IMF7	2	2000	3.6278e-1(5.34e-3)	5.3313e-1(4.88e-3)	3.0399e-1(4.83e-3)	2.7832e-1(6.00e-17)	5.1928e-01(9.61e-03)	6.8488e-01(3.67e-03)	4.5665e-2 (1.07e-2) -	1.8225e-2(2.23e-3)
IMF7	2	5000	3.8990e-1(2.86e-3)	5.3909e-1(5.31e-3)	3.0278e-1(1.88e-3)	2.5266e-2(7.49e-18)	5.2601e-01(1.25e-02)	6.8247e-01(1.41e-03)	3.9243e-2 (1.36e-2) -	1.7306e-2(3.61e-3)
IMF8	3	500	7.7309e+0(1.62e+0)	2.5848e+1(1.17e+0)	5.5544e+0(1.78e-1)	5.8663e-1(0.00e+0)	2.1968e+01(2.08e+00)	2.5519e+00(3.31e-01)	4.9722e-1 (2.33e-1) =	3.6191e-1(1.50e-3)
IMF8	3	1000	1.7483e+1(2.51e+0)	5.3550e+1(1.30e+0)	1.0744e+1(5.58e-1)	5.5627e-1(1.20e-16)	4.8125e+01(2.20e+00)	4.6656e+00(9.85e-01)	3.6061e-1(2.14e-4) +	3.6172e-1(5.83e-4)
IMF8	3	2000	6.8070e+1(1.84e+1)	1.0897e+2(3.42e+0)	2.1241e+1(6.17e-1)	5.7880e-1(0.00e+0)	9.8647e+01(2.02e+00)	1.0690e+01(2.99e-01)	4.5153e-1 (2.41e-1) -	3.6164e-1(6.30e-4)
IMF8	3	5000	2.8946e+2(5.34e+1)	2.7392e+2(2.80e+0)	5.3796e+1(1.01e+0)	6.7750e-1(0.00e+0)	2.5310e+02(4.94e+00)	1.9105e+01(7.87e+00)	4.9794e-1 (2.34e-1) =	3.6143e-1(3.17e-4)
IMF9	2	500	3.4468e-1(1.69e-2)	7.2260e-1(7.47e-3)	6.6999e-1(7.13e-3)	3.7396e-1(6.00e-17)	6.8698e-01(6.54e-03)	4.5716e-01(3.28e-02)	1.6097e-1 (1.23e-1) -	6.8946e-2(9.76e-3)
IMF9	2	1000	4.0239e-1(1.18e-2)	9.1184e-1(8.53e-3)	8.1138e-1(1.24e-2)	4.6196e-1(0.00e+0)	8.6745e-01(2.38e-02)	5.1276e-01(2.67e-02)	8.8517e-2 (1.12e-2) -	7.6492e-2(1.50e-2)
IMF9	2	2000	5.0773e-1(1.62e-2)	1.3250e+0(2.02e-2)	1.1172e+0(1.54e-2)	5.0605e-1(1.20e-16)	1.2498e+00(3.01e-02)	6.4376e-01(1.27e-02)	1.3462e-1 (1.46e-2) -	9.9996e-2(3.17e-2)
IMF9	2	5000	7.9123e-1(1.50e-2)	2.8203e+0(4.39e-2)	2.1071e+0(4.47e-2)	3.5713e-2(0.00e+0) +	2.5414e+00(5.64e-02)	8.9786e-01(2.41e-02)	2.4210e-1 (9.26e-2) -	1.5646e-1(4.45e-2)
+/-/=			0/36/0	0/36/0	0/36/0	3/30/3	0/36/0	0/36/0	3/25/8	

Table A. 6: The HV results of CL-CSO and its seven compared algorithms on solving IMF test problems with two objectives and 500-5000 decision variables.

Problem	M	D	FDV	LMEA	LMOCSO	S-ECSO	LSMOEA/D	CCSO	NN-CSO	CL-CSO
IMF1	2	500	1.6174e-1 (2.34e-2) -	0.0000e+0 (0.00e+0) -	0.0000e+0 (0.00e+0) -	3.0154e-1 (0.00e+0) -	0.0000e+0(0.00e+0)-	0.0000e+0(0.00e+0)-	3.9733e-1(6.22e-2)=	4.4269e-1 (1.97e-2)
IMF1	2	1000	6.4726e-2 (2.33e-2) -	0.0000e+0 (0.00e+0) -	0.0000e+0 (0.00e+0) -	2.9254e-1 (0.00e+0) -	0.0000e+0(0.00e+0)-	0.0000e+0(0.00e+0)-	3.5294e-1(8.71e-2)-	4.6327e-1 (3.11e-2)
IMF1	2	2000	1.9480e-2 (1.08e-2) -	0.0000e+0 (0.00e+0) -	0.0000e+0 (0.00e+0) -	2.9400e-1 (0.00e+0) -	0.0000e+0(0.00e+0)-	0.0000e+0(0.00e+0)-	3.8762e-1(1.44e-2)-	4.5788e-1 (2.21e-2)
IMF1	2	5000	0.0000e+0 (0.00e+0) -	0.0000e+0 (0.00e+0) -	0.0000e+0 (0.00e+0) -	3.0179e-1 (0.00e+0) -	0.0000e+0(0.00e+0)-	0.0000e+0(0.00e+0)-	2.9489e-1(5.59e-2)-	4.5629e-1 (1.30e-2)
IMF2	2	500	0.0000e+0 (0.00e+0) -	0.0000e+0 (0.00e+0) -	0.0000e+0 (0.00e+0) -	9.0909e-2 (0.00e+0) +	0.0000e+0(0.00e+0)-	0.0000e+0(0.00e+0)-	2.4977e-2(3.32e-2)-	9.0781e-2 (2.48e-4)
IMF2	2	1000	0.0000e+0 (0.00e+0) -	0.0000e+0 (0.00e+0) -	0.0000e+0 (0.00e+0) -	9.0909e-2 (0.00e+0) +	0.0000e+0(0.00e+0)-	0.0000e+0(0.00e+0)-	2.2897e-2(2.86e-2)-	9.0909e-2 (3.68e-8)
IMF2	2	2000	0.0000e+0 (0.00e+0) -	0.0000e+0 (0.00e+0) -	0.0000e+0 (0.00e+0) -	9.0909e-2 (0.00e+0) =	0.0000e+0(0.00e+0)-	0.0000e+0(0.00e+0)-	1.6165e-2(2.85e-2)-	9.0909e-2 (1.97e-16)
IMF2	2	5000	0.0000e+0 (0.00e+0) -	0.0000e+0 (0.00e+0) -	0.0000e+0 (0.00e+0) -	9.0909e-2 (0.00e+0) =	0.0000e+0(0.00e+0)-	0.0000e+0(0.00e+0)-	1.5911e-2(3.16e-2)-	9.0909e-2 (2.10e-16)
IMF3	2	500	0.0000e+0 (0.00e+0) -	0.0000e+0 (0.00e+0) -	0.0000e+0 (0.00e+0) -	2.3677e-1 (0.00e+0) -	0.0000e+0(0.00e+0)-	0.0000e+0(0.00e+0)-	3.4232e-1(4.74e-2)-	3.7826e-1 (2.57e-3)
IMF3	2	1000	0.0000e+0 (0.00e+0) -	0.0000e+0 (0.00e+0) -	0.0000e+0 (0.00e+0) -	2.2319e-1 (0.00e+0) -	0.0000e+0(0.00e+0)-	0.0000e+0(0.00e+0)-	2.6406e-1(1.07e-1)-	3.7994e-1 (1.29e-3)
IMF3	2	2000	0.0000e+0 (0.00e+0) -	0.0000e+0 (0.00e+0) -	0.0000e+0 (0.00e+0) -	2.0295e-1 (0.00e+0) -	0.0000e+0(0.00e+0)-	0.0000e+0(0.00e+0)-	3.2552e-1(6.00e-2)-	3.8047e-1 (9.63e-4)
IMF3	2	5000	0.0000e+0 (0.00e+0) -	0.0000e+0 (0.00e+0) -	0.0000e+0 (0.00e+0) -	3.2598e-1 (6.00e-17) -	0.0000e+0(0.00e+0)-	0.0000e+0(0.00e+0)-	3.4809e-1(6.27e-2)-	3.8015e-1 (1.23e-3)
IMF4	3	500	0.0000e+0 (0.00e+0) -	0.0000e+0 (0.00e+0) -	0.0000e+0 (0.00e+0) -	2.2875e-1 (6.00e-17) =	0.0000e+0(0.00e+0)-	0.0000e+0(0.00e+0)-	1.2784e-1(1.61e-1)=	2.3703e-1 (1.36e-1)
IMF4	3	1000	0.0000e+0 (0.00e+0) -	0.0000e+0 (0.00e+0) -	0.0000e+0 (0.00e+0) -	3.0069e-1 (0.00e+0) +	0.0000e+0(0.00e+0)-	0.0000e+0(0.00e+0)-	4.4246e-2(1.17e-1)-	2.8988e-1 (6.74e-2)
IMF4	3	2000	0.0000e+0 (0.00e+0) -	0.0000e+0 (0.00e+0) -	0.0000e+0 (0.00e+0) -	2.3927e-1 (0.00e+0) -	0.0000e+0(0.00e+0)-	0.0000e+0(0.00e+0)-	4.5971e-2(1.22e-1)-	3.1537e-1 (3.25e-3)
IMF4	3	5000	0.0000e+0 (0.00e+0) -	0.0000e+0 (0.00e+0) -	0.0000e+0 (0.00e+0) -	2.8726e-1 (6.00e-17) -	0.0000e+0(0.00e+0)-	0.0000e+0(0.00e+0)-	4.6023e-2(1.22e-1)-	3.1760e-1 (1.16e-3)
IMF5	2	500	4.3442e-1 (4.40e-3) -	2.9975e-1 (4.75e-3) -	4.3318e-1 (1.78e-3) -	2.8512e-1 (0.00e+0) -	3.2712e-1(7.84e-3)-	4.0959e-1(3.71e-3)-	6.5351e-1(6.95e-3)-	6.7139e-1 (2.58e-3)
IMF5	2	1000	4.0745e-1 (8.11e-3) -	2.9818e-1 (2.30e-3) -	4.3446e-1 (1.49e-3) -	2.9256e-1 (0.00e+0) -	3.1296e-1(8.00e-3)-	4.1307e-1(4.17e-3)-	6.6126e-1(3.29e-3)-	6.6892e-1 (2.46e-3)
IMF5	2	2000	3.7850e-1 (4.89e-3) -	2.9565e-1 (1.59e-3) -	4.3291e-1 (3.37e-3) -	2.8394e-1 (0.00e+0) -	3.0415e-1(5.07e-3)-	4.1752e-1(5.63e-3)-	6.6437e-1(3.61e-3)-	6.6947e-1 (1.57e-3)
IMF5	2	5000	3.6287e-1 (5.04e-3) -	2.9114e-1 (1.08e-3) -	4.3332e-1 (1.05e-3) -	2.9396e-1 (0.00e+0) -	2.9965e-1(7.58e-3)-	4.1886e-1(3.07e-3)-	6.6797e-1(2.58e-3)=	6.6951e-1 (6.46e-4)
IMF6	2	500	1.1156e-1 (1.17e-2) -	3.5818e-3 (5.07e-3) -	1.9849e-1 (4.76e-3) -	9.0909e-2 (0.00e+0) -	2.9084e-2(9.54e-3)-	1.6149e-1(7.45e-3)-	3.6531e-1(2.35e-3)-	3.7290e-1 (1.39e-3)
IMF6	2	1000	8.7070e-2 (5.92e-3) -	9.2566e-4 (2.23e-3) -	2.0348e-1 (1.95e-3) -	9.0909e-2 (0.00e+0) -	1.4865e-2(8.41e-3)-	1.8203e-1(4.82e-3)-	3.6976e-1(1.85e-3)-	3.7342e-1 (1.17e-3)
IMF6	2	2000	6.1278e-2 (4.38e-3) -	0.0000e+0 (0.00e+0) -	2.0343e-1 (2.00e-3) -	9.0909e-2 (0.00e+0) -	1.8594e-3(2.61e-3)-	1.8954e-1(3.68e-3)-	3.7227e-1(1.64e-3)=	3.7305e-1 (7.09e-4)
IMF6	2	5000	3.7299e-2 (9.35e-3) -	0.0000e+0 (0.00e+0) -	2.0329e-1 (5.60e-4) -	9.0909e-2 (0.00e+0) -	0.0000e+0(0.00e+0)-	1.9329e-1(3.06e-3)-	3.7197e-1(1.36e-3)=	3.1962e-1 (1.41e-1)
IMF7	2	500	1.0920e-1 (4.96e-3) -	2.6297e-2 (2.96e-3) -	1.2728e-1 (3.01e-3) -	2.4156e-1 (3.00e-17) -	4.3466e-2(3.94e-3)-	1.1851e-1(1.60e-3)-	3.2990e-1(4.69e-2)-	3.6690e-1 (8.46e-3)
IMF7	2	1000	9.4386e-2 (5.98e-3) -	2.5237e-2 (8.81e-4) -	1.2822e-1 (1.79e-3) -	2.2316e-1 (3.00e-17) -	3.1003e-2(4.56e-3)-	1.2353e-1(1.73e-3)-	3.6851e-1(2.18e-2)-	3.6674e-1 (8.27e-3)
IMF7	2	2000	8.6133e-2 (3.18e-3) -	2.3691e-2 (1.18e-3) -	1.2832e-1 (8.44e-4) -	1.7460e-1 (0.00e+0) -	2.6958e-2(2.79e-3)-	1.2531e-1(2.31e-3)-	3.7221e-1(1.10e-2)-	3.6979e-1 (2.11e-3)
IMF7	2	5000	7.2520e-2 (1.24e-3) -	2.2436e-2 (9.69e-4) -	1.2719e-1 (1.49e-3) -	3.6346e-1 (6.00e-17) -	2.5347e-2(3.81e-3)-	1.2668e-1(1.29e-3)-	3.7886e-1(1.37e-2)-	3.7092e-1 (4.06e-3)
IMF8	3	500	0.0000e+0 (0.00e+0) -	0.0000e+0 (0.00e+0) -	0.0000e+0 (0.00e+0) -	2.5250e-1 (0.00e+0) -	0.0000e+0(0.00e+0)-	0.0000e+0(0.00e+0)-	2.9841e-1(1.42e-1)=	3.7650e-1 (3.57e-3)
IMF8	3	1000	0.0000e+0 (0.00e+0) -	0.0000e+0 (0.00e+0) -	0.0000e+0 (0.00e+0) -	2.8244e-1 (6.00e-17) -	0.0000e+0(0.00e+0)-	0.0000e+0(0.00e+0)-	3.8295e-1(1.85e-3)+	3.7646e-1 (2.92e-3)
IMF8	3	2000	0.0000e+0 (0.00e+0) -	0.0000e+0 (0.00e+0) -	0.0000e+0 (0.00e+0) -	2.3927e-1 (0.00e+0) -	0.0000e+0(0.00e+0)-	0.0000e+0(0.00e+0)-	3.2914e-1(1.44e-1)-	3.7689e-1 (2.94e-3)
IMF8	3	5000	0.0000e+0 (0.00e+0) -	0.0000e+0 (0.00e+0) -	0.0000e+0 (0.00e+0) -	1.7355e-1 (0.00e+0) -	0.0000e+0(0.00e+0)-	0.0000e+0(0.00e+0)-	2.9565e-1(1.50e-1)=	3.7572e-1 (2.18e-3)
IMF9	2	500	3.0345e-1 (1.56e-2) -	5.8893e-2 (1.97e-3) -	8.6108e-2 (3.42e-3) -	3.8136e-1 (6.00e-17) -	7.2623e-2(3.00e-3)-	7.0207e-2(3.95e-3)-	5.3977e-1(1.29e-1)-	6.4279e-1 (1.00e-2)
IMF9	2	1000	2.4952e-1 (1.07e-2) -	7.4709e-3 (1.46e-3) -	3.1503e-2 (3.29e-3) -	3.6400e-1 (0.00e+0) -	1.6833e-2(5.78e-3)-	2.0041e-2(2.85e-3)-	6.2224e-1(1.29e-2)=	6.3413e-1 (1.71e-2)
IMF9	2	2000	1.6718e-1 (9.68e-3) -	0.0000e+0 (0.00e+0) -	0.0000e+0 (0.00e+0) -	3.4748e-1 (6.00e-17) -	0.0000e+0(0.00e+0)-	0.0000e+0(0.00e+0)-	5.9318e-1(1.66e-2)=	6.1073e-1 (3.25e-2)
IMF9	2	5000	3.5139e-2 (3.98e-3) -	0.0000e+0 (0.00e+0) -	0.0000e+0 (0.00e+0) -	6.8150e-1 (1.20e-16) +	0.0000e+0(0.00e+0)-	0.0000e+0(0.00e+0)-	5.0910e-1(5.92e-2)=	5.5884e-1 (3.96e-2)
+/-/=			0/36/0	0/36/0	0/36/0	4/29/3	0/36/0	0/36/0	1/25/10	

Table A. 7: The average IGD values of CL-CSO with different values of p in solving IMF problems with 500 decision variables.

Problem	M	D	$p=0$	$p=0.3$	$p=0.6$	$p=1$	$p=0.9$
IMF1	2	500	4.04E-01	3.73E-01	3.49E-01	4.83E+00	3.80E-01
IMF2	2	500	6.09E-01	6.09E-01	6.09E-01	7.03E+00	6.09E-01
IMF3	2	500	2.29E+00	9.52E-03	9.14E-03	8.23E+00	1.06E-02
IMF4	3	500	4.66E+01	5.45E-01	5.11E+01	4.97E+02	5.85E-01
IMF5	2	500	5.20E-02	4.88E-02	4.67E-02	2.00E-01	4.48E-02
IMF6	2	500	9.37E-02	9.34E-02	9.18E-02	2.81E-01	9.16E-02
IMF7	2	500	7.18E-02	7.58E-02	2.53E-02	3.04E-01	2.07E-02
IMF8	3	500	3.61E-01	3.61E-01	3.61E-01	4.49E+00	3.62E-01
IMF9	2	500	7.56E-02	7.28E-02	8.46E-02	6.45E-01	6.89E-02
Friedman Rank			3.2	2.4	2.2	5	2

Table A. 8: The average IGD values of CL-CSO with different values of T in solving IMF problems with 500 decision variables.

Problem	M	D	$T=0$	$T=5$	$T=20$	$T=35$	$T=50$	$T=10$
IMF1	2	500	4.39E-01	3.84E-01	3.61E-01	3.70E-01	3.91E-01	3.80E-01
IMF2	2	500	6.09E-01	6.10E-01	6.09E-01	6.13E-01	6.10E-01	6.09E-01
IMF3	2	500	1.24E-02	9.81E-03	1.05E-02	9.56E-03	9.89E-03	1.06E-02
IMF4	3	500	9.43E+01	3.56E+00	5.49E-01	3.22E+00	2.71E+01	5.85E-01
IMF5	2	500	4.85E-02	4.59E-02	4.54E-02	4.76E-02	4.84E-02	4.48E-02
IMF6	2	500	8.95E-02	9.03E-02	9.02E-02	9.10E-02	9.25E-02	9.16E-02
IMF7	2	500	2.64E-02	2.01E-02	2.24E-02	2.75E-02	2.45E-02	2.07E-02
IMF8	3	500	3.62E-01	3.62E-01	3.62E-01	3.62E-01	3.62E-01	3.62E-01
IMF9	2	500	7.45E-02	6.22E-02	6.84E-02	6.54E-02	8.14E-02	6.89E-02
Friedman Rank			4.5	2.9	2.4	3.5	4.7	3